is that your application handles these errors gracefully, providing a suitable response to the user and not causing your whole application to fail.

The design philosophy for ASP.NET Core is that every feature is opt-in. So because error handling is a feature, you need to enable it explicitly in your application. Many types of errors could occur in your application, and you have many ways to handle them, but in this section I focus on a single type of error: exceptions.

Exceptions typically occur whenever you find an unexpected circumstance. A typical (and highly frustrating) exception you'll no doubt have experienced before is `NullReferenceException`, which is thrown when you attempt to access a variable that hasn't been initialized.[3] If an exception occurs in a middleware component, it propagates up the pipeline, as shown in figure 4.14. If the pipeline doesn't handle the exception, the web server returns a 500 status code to the user.